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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB

RECORD

2026

SKILL 1

- **Create an IAM user with programmatic access only, generate and download the Access Key ID and Secret Access Key, and configure an AWS billing alert to receive an email notification when the Free Tier usage approaches or exceeds the specified limit.**
- **IAM User Creation, Access Key Generation and AWS Billing Alert Configuration**

Description:

- Create an IAM user in AWS.
- Generate Access Key ID and Secret Access Key.
- Download the access keys.
- Enable AWS Free Tier usage alerts.
- Create a monthly AWS budget.
- Configure email notifications for billing alerts.
- Monitor AWS usage and avoid unexpected charges.

Step 1: Open your browser and go to:

AWS Management Console

Step 2: Log in using your AWS root account credentials.

Step 3: At the top search bar, type `IAM`.

Step 4: Select IAM (Identity and Access Management).

Step 5: On the left side, click Users.

Step 6: Click Create user.

Step 7: Enter a username.

For example:

```
student1
```

Step 8: Do not enable AWS Management Console access

.Step 9: Click Next.

Step 10: Under Set permissions, select:

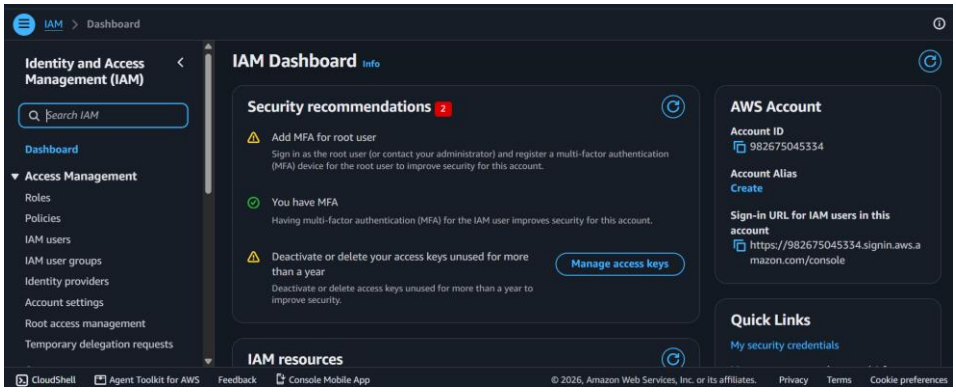
`Attach policies directly`

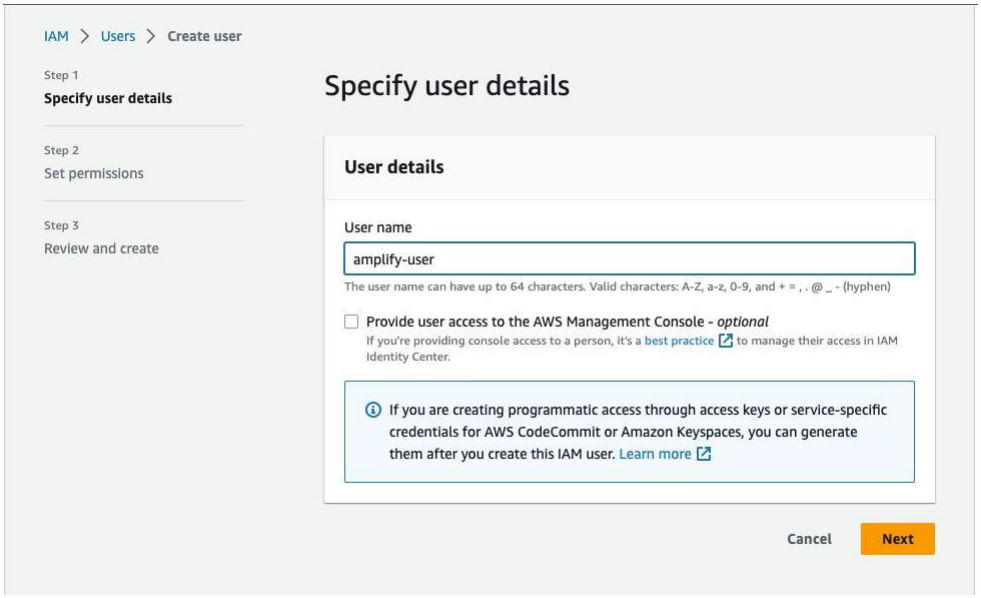
Step 11: Select the required permission policy.

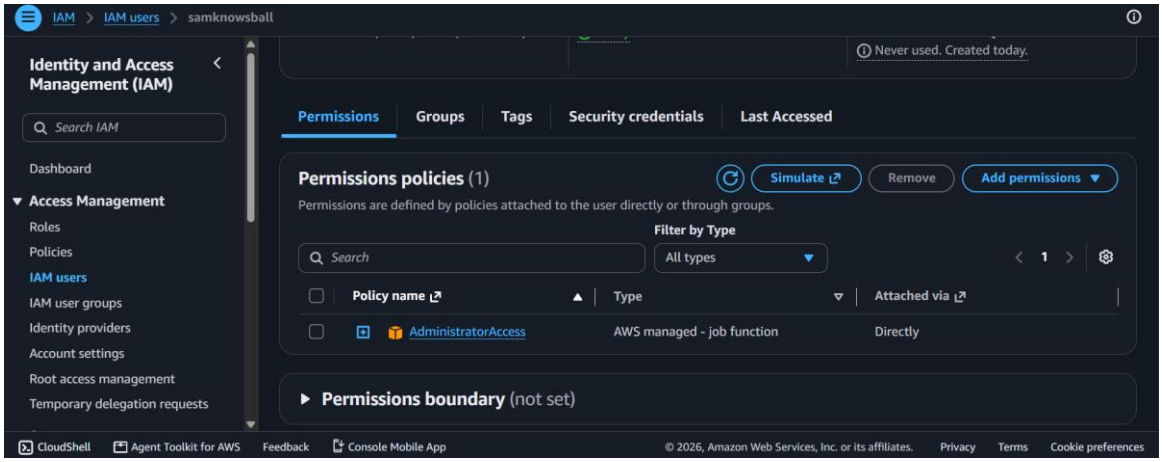
Step 12: Click Next.

Step 13: Review configuration.

Step 14: Click Create user.







Generate and Download Access Keys

Step 1: Go to IAM.

Step 2: Click Users.

Step 3: Click the username created:

`Student-IAM-User`

Step 4: Open the Security credentials tab.

Step 5: Scroll down until we find the Access keys section.

Step 6: Click Create access key.

Step 7: Select the use case:

`Command Line Interface (CLI)`

Step 8: Acknowledge the recommendation and click Next.

Step 9: Enter a description tag (optional).

Example:

`Student-CLI-Access-Key`

Step 10: Click Create access key.

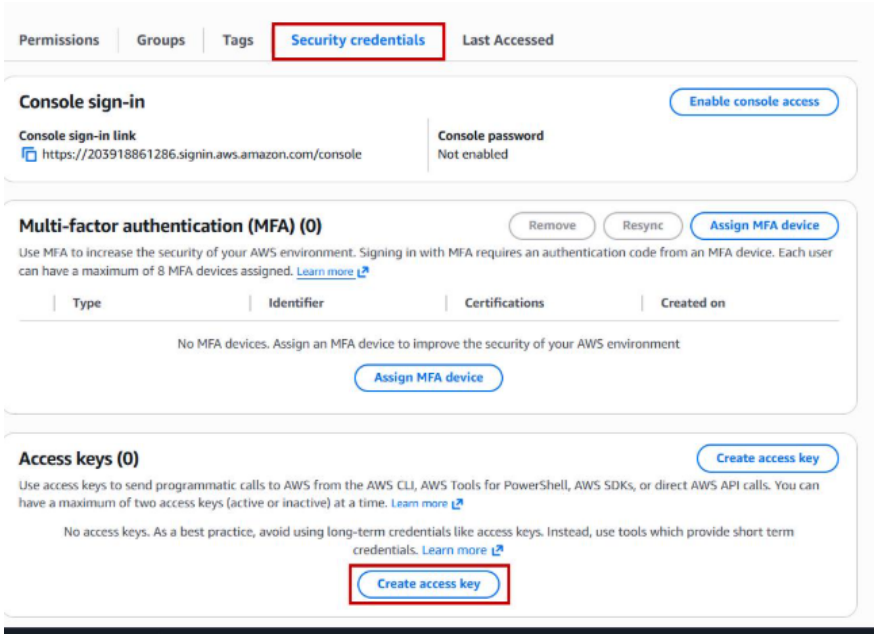
Now AWS will display two important values:

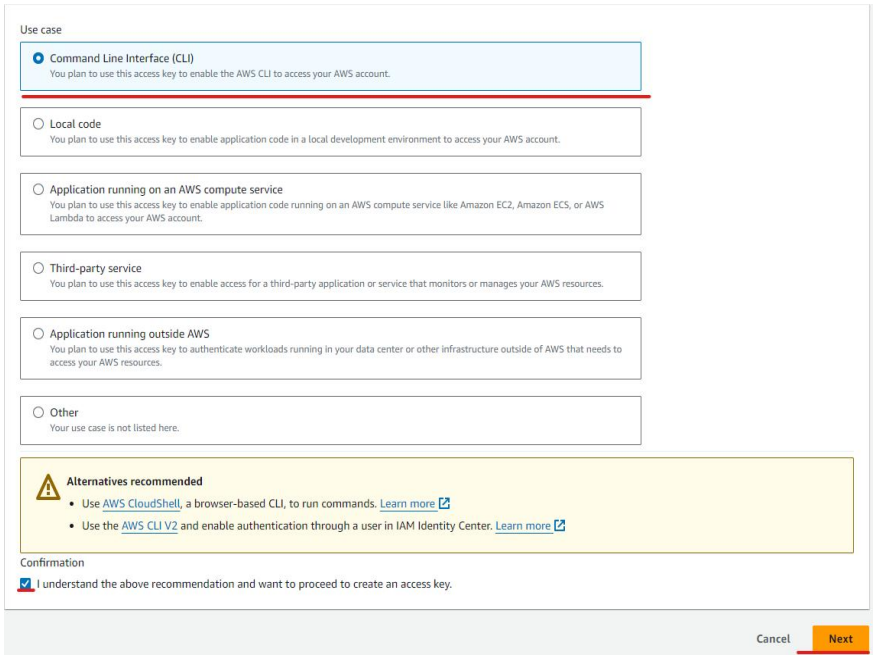
1. Access Key ID
2. Secret Access Key

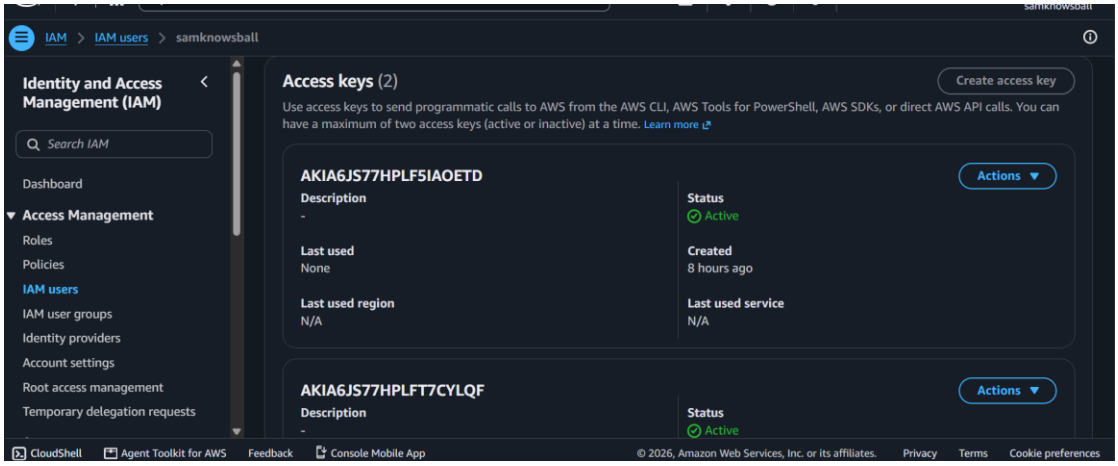
Step 11: Click Download .csv file.

Step 12: Open the downloaded CSV file and check that both credentials are present.

Step 13: Click Done.







Configure AWS Free Tier Usage Alerts

Step 1: Log in using AWS root account.

Step 2: In the AWS search bar, type:

`Billing and Cost Management`

Step 3: Open Billing and Cost Management.

Step 4: On the left navigation panel, find Preferences.

Step 5: Click Billing preferences.

Step 6: Find the section called Alert preferences.

Step 7: Click Edit.

Step 8: Find this checkbox:

`Receive AWS Free Tier usage alerts`

Step 9: Enable the checkbox.

Step 10: Find:

`Additional email address to receive alerts`

Enter your email address if required.

Step 11: Click Update.

Step 12: Verify that the Free Tier usage alert option is enabled.

Configure AWS Billing Alerts Using Custom Budget

Step 1: Open AWS Billing and Cost Management.

Step 2: On the left panel, find Budgets under Budgets and Planning.

Step 3: Click Budgets.

Step 4: Click Create budget.

Step 5: Under Budget setup, select:

`Use a template`

Step 6: Under Templates, select:

`Zero spend budget`

This template notifies you when your spending exceeds $0.01.

Step 7: Enter a budget name.

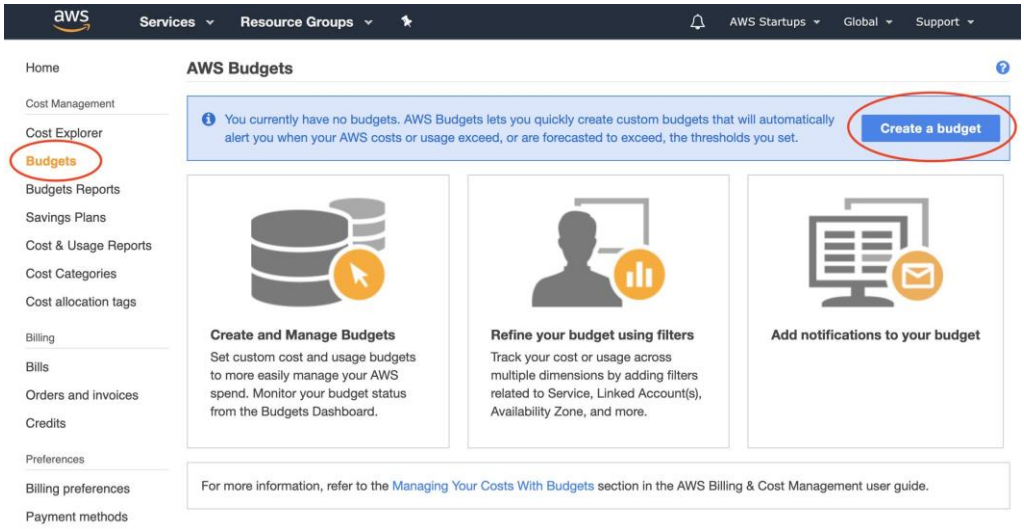
Example:

`My-Zero-Spend-Budget`

Step 8: Enter email address.

Step 9: Follow the remaining confirmation steps and create the budget.

Step 10: Check whether the budget appears in the AWS Budgets dashboard.



Set your budget

Set your budget details, including your budgeted amount. From there, you can refine your budget using the optional budget parameters.

Budget details

Name

AWS Total Monthly Cost Demo Budget

Period

Monthly

Budget effective dates

Recurring budgets will renew on the first day of every monthly billing period. Expiring budgets will stop renewing on the last day of the expiration month.

- (•) Recurring Budget
- ( ) Expiring Budget

Start Month

Nov 2019

Budget amount

- (•) **Fixed** — Create a budget that tracks against a single monthly budgeted amount.
- ( ) **Monthly Budget Planning** — Specify your budgeted amount for each budget period.

Budgeted amount

$200

Last month's cost $176.07

▼ Alert #1

Remove

Set alert threshold

Threshold
When should this alert be triggered?

80.00 | % of budgeted amount

Trigger
How should this alert be triggered?

Actual

Summary: When your actual cost is greater than **80.00% ($80.00)** of your **budgeted amount ($100.00)**, the alert threshold will be exceeded.

Notification preferences

Select one or more notification preferences to receive alerts.

Email recipients
Specify the email recipients you want to notify when the threshold has exceeded.

firstname.lastname@colorado.edu

Maximum number of email recipients is 10.

▶ **Amazon SNS Alerts - *Optional*** Info

▶ **AWS Chatbot Alerts**

+ Add alert threshold

Cancel | Previous | Next

Choose budget type Info

Budget setup

- **Use a template (simplified)** — Use the recommended configurations. You can change some configuration options after the budget is created.
- **Customize (advanced)** — Customize a budget to set parameters specific to your use case. You can customize the time period, the start month, and specific accounts.

Templates - *new*

Choose a template that best matches your use case.

- **Zero spend budget** — Create a budget that notifies you once your spending exceeds $0.01 which is above the AWS Free Tier limits.
- **Monthly cost budget** — Create a monthly budget that notifies you if you exceed, or are forecasted to exceed, the budget amount.
- **Daily Savings Plans coverage budget** — Create a coverage budget for your Savings Plans that notifies you when you fall below the defined target.
- **Daily reservation utilization budget** — Create a utilization budget for your reservations that notifies you when you fall below the defined target.

Review

Budget details

Name	Start date	Budget amount
Monthly-budget	Jun 2023	$5.00
Period	**End date**	
Monthly	-	

▶ Additional budget parameters

Step 3: Configure alerts — Edit

Alerts

Alert #1

Threshold
80% of budgeted amount

Threshold measured against
Actual costs

Step 4: Attach actions - optional — Edit

Actions

You have no budgets actions

Cancel | Previous | Create budget